

Two-Step Equations in ContextAdditional Practice 7A.01.04 **ANSWER KEY**

Lisa solved the equation $-2x - 3 = 5$. Her work is shown. Use it for question 1.

$$\begin{aligned} -2x - 3 + 3 &= 5 + 3 \\ -2x &= 8 \\ -2x \div (-2) &= 8 \div (-2) \\ x &= 4 \end{aligned}$$

1. Use substitution to decide whether Lisa's solution is correct. If it is not, describe her error and solve the equation correctly.

$-2(4) - 3 = -8 - 3 = -11$, and $-11 \neq 5$, so it is not correct.

Her error: $8 \div (-2)$ is -4 , not 4. She dropped the negative sign when she divided.

$x = -4$ Check: $-2(-4) - 3 = 8 - 3 = 5$ ✓

Zoey solved the equation $3m + 5 = 8$. Her work is shown. Use it for question 2.

$$\begin{aligned} 3m + 5 - 5 &= 8 - 5 \\ 3m &= -3 \\ 3m \div 3 &= -3 \div 3 \\ m &= -1 \end{aligned}$$

2. Use substitution to decide whether Zoey's solution is correct. If it is not, describe her error and solve the equation correctly.

$3(-1) + 5 = -3 + 5 = 2$, and $2 \neq 8$, so it is not correct.

Her error: $8 - 5$ is 3, not -3 . The subtraction itself was done wrongly.

$m = 1$ Check: $3(1) + 5 = 3 + 5 = 8$ ✓

For questions 3 through 5, solve the equation, then say what the solution means in the situation.

3. A taxi charges \$1.75 plus \$0.75 for each mile traveled. Your trip cost \$14.50. The equation $1.75 + 0.75x = 14.50$ represents the situation.

$0.75x = 12.75$ $x = 17$

The trip was 17 miles.

Check: $1.75 + 0.75(17) = 1.75 + 12.75 = 14.50$.

4. A zipline course charges a flat rate of \$75 plus \$5.50 for each line you cross. One person's total was \$124.50. The equation $75 + 5.50z = 124.50$ represents the situation.

$$5.50z = 49.50 \quad z = 9$$

That person crossed 9 lines.

5. A runner warms up for 15 minutes and then runs each mile in 6 minutes. The total time is 45 minutes. The equation $15 + 6m = 45$ represents the situation.

$$6m = 30 \quad m = 5$$

The runner ran 5 miles.

For questions 6 through 10, solve the equation and check your solution by substitution.

6. $125 = 15 + 5x$

$$125 - 15 = 15 + 5x - 15$$

$$110 = 5x$$

$$110 \div 5 = 5x \div 5$$

$$22 = x$$

$$\text{Check: } 15 + 5(22) = 15 + 110 = 125 \checkmark$$

Subtraction property, then division property. The variable ends on the right; that is the symmetric property and it is finished.

7. $9.2 = 6.1 + \frac{h}{6}$

$$9.2 - 6.1 = 6.1 + h/6 - 6.1$$

$$3.1 = h/6$$

$$3.1 \times 6 = (h/6) \times 6$$

$$18.6 = h$$

$$\text{Check: } 6.1 + 18.6 \div 6 = 6.1 + 3.1 = 9.2 \checkmark$$

Subtraction property, then multiplication property.

8. $17.5c - 15 = 98.75$

$$17.5c = 113.75$$

$$c = 6.5$$

$$\text{Check: } 17.5(6.5) - 15 = 113.75 - 15 = 98.75 \checkmark$$

Addition property, then division property.

9. $\frac{t - 50}{4} = 400$

$$(t - 50)/4 \times 4 = 400 \times 4$$

$$t - 50 = 1600$$

$$t = 1650$$

$$\text{Check: } (1650 - 50) \div 4 = 1600 \div 4 = 400 \checkmark$$

Multiplication property, then addition property. Here the whole numerator is divided by 4, so undo the division first.

10. $16.42 + 2.51r = 26.46$

$$2.51r = 10.04$$

$$r = 4$$

$$\text{Check: } 16.42 + 2.51(4) = 16.42 + 10.04 = 26.46 \checkmark$$

Subtraction property, then division property.